

Coding & Solution

Date	29 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below

Solution Summary

Field	Details
Repository Link / URL	https://github.com/yasminshaik18/Flood_prediction.git
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost
Key Features Implemented	1. Data preprocessing and cleaning of flood prediction dataset 2. Training of four ML classifiers — Decision Tree, Random Forest, KNN, and XGBoost

	3. Model comparison and automatic selection of best performing model (XGBoost — 81.60% accuracy) 4. Saved trained model using joblib for reuse 5. Flask web application with input form for 20 environmental parameters 6. Real-time flood risk prediction — Flood Likely or No Flood 7. Risk level indicator bar on result page 8. Actionable recommendations based on prediction result 9. About page with model performance comparison table 10. Responsive and professional UI with navigation bar
Pending / Incomplete Features	1. User authentication and login system 2. Prediction history page to store past predictions 3. Real-time weather API integration for auto-filling input fields 4. SMS or email alert system for flood warnings
Setup / Run Instructions	1. Clone the repository from GitHub 2. Install required libraries using: pip install -r requirements.txt 3. Train the model by running: python train_model.py 4. Start the Flask web application using: python app.py 5. Open browser and go to: http://127.0.0.1:5000 6. Enter values from 1 to 10 for all 20 environmental parameters and click Predict Flood Risk

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The flood prediction system was built using a real-world dataset containing over 1 million records sourced from Kaggle. Four machine learning classifiers were trained and evaluated. XGBoost achieved the highest accuracy of 81.60% and was selected as the deployed model. The Flask web application provides an intuitive interface for disaster management authorities to assess flood risk in real time by entering current environmental parameters.